GinCol Lab · Aula

Student worksheet

Physics and Chemistry · 1st year of Bachillerato · The atom and the periodic table · Class 3 of 6 · With Abril

The light that gives each element away

Class video: https://youtu.be/SfjZTFB6QJQ

The code leads to the video.

Name: _____ Date: _____

Before watching

1. In the previous class you saw the flames of lithium, sodium and potassium. What colour was each one? What was odd about it, if they belong to the same family?

2. If you pass the light of an old-style light bulb through a prism, a rainbow comes out. What do you think will come out with the light of a hydrogen tube that glows because a current passes through it? Draw it and explain why you think so.

3. A nanometre is 10^{-9} m. Write 656.3 nm in metres, in scientific notation: 656.3 nm = _____ × 10^{-9} m = _____ × 10^{-7} m.

While you watch the video

1. What did Auguste Comte write in 1835 about the stars? Why did it seem reasonable?

2. What is wavelength? And frequency? If the wavelength gets shorter, what happens to the frequency? Why?

3. What is the difference between a continuous spectrum and an emission spectrum? Which elements were discovered by their lines?

4. What did Max Planck propose? What did Albert Einstein add?

5. **The fill-in-the-gaps example.** Hydrogen's blue-green line:

   λ = 486.1 nm = _____ × 10^{-7} m

   E = (6.63 × 10^{-34} × 3.00 × 10^{8}) / _____ = _____ × 10^{-19} J

6. **Before Abril tells you**: a very intense red light and a very faint violet one, which carries photons with more energy? Why?

7. Why could Rutherford's atom not explain the lines? What is a postulate?

8. Hydrogen has only one electron. Why does it have so many rungs? How many of them is the electron on at once?

9. **Hydrogen's ladder.** Fill in the energy of each rung, in 10^{-19} J.

n = 1	n = 2	n = 3	n = 4	n = 5	n = 6

10. Why do the rungs carry a minus sign? Where did Bohr get the energy needed to pull the electron away?

11. **Your prediction.** What colour will the line from the jump from 6 to 2 be? Why?

12. **The test.** Complete the table.

Jump	Photon energy (10^{-19} J)	Colour	Wavelength (nm)
From 3 to 2			
From 4 to 2			
From 5 to 2			
From 6 to 2			

13. With which atom does Bohr's model stop working? Why do you think it is precisely that one?

14. Why do lithium, sodium and potassium give different colours? What do they have in common?

15. On the whiteboard, the jumps of their ladders are drawn to scale. Which of the red and yellow jumps is the biggest? Does it fit the colour of its line? And potassium's violet?

16. Why does a line spectrum work as an element's fingerprint, and a continuous one does not?

17. What is an absorption spectrum? What is needed to see one?

18. What did Fraunhofer see? What did Kirchhoff do, and what did he conclude?

19. How and where was helium discovered? Was it a dark line or a bright one?

20. What did Cecilia Payne calculate? What did she herself write about her result?

In class: redo Bohr's test and read the light of the Sun

Part A: redo Bohr's test. Your group takes one jump. Data: the energy of rung n is $E_n = -21.8 \times 10^{-19}$ J / n^2; $h \cdot c = 6.63 \times 10^{-34} \times 3.00 \times 10^{8} = 1.989 \times 10^{-25}$ J·m; and the wavelength of the photon is $\lambda = h \cdot c / \Delta E$.

Our jump	Upper E (10^{-19} J)	Lower E (10^{-19} J)	ΔE (10^{-19} J)	λ (m)	λ (nm)	Colour
From __ to 2						

What is measured (NIST, in air): 656.28 nm, 486.13 nm, 434.05 nm and 410.17 nm.

1. Does your result match the measurement? How far off is it, in nanometres and as a percentage?

2. If your result does not match exactly, does the model fail? Redo the calculation with more figures: $h = 6.626 \times 10^{-34}$ J·s, $c = 2.998 \times 10^{8}$ m/s and -21.79×10^{-19} J on rung 1. What changes?

3. What would have to come out for Bohr's model to be refuted?

Part B: read the light of the Sun. These are some of the most marked dark lines in the Sun's spectrum. Below them, the lines of some elements, measured in the laboratory. All of them in nanometres.

Dark line of the Sun	**422.7**	**438.4**	**486.1**	**516.7**	**517.3**	**518.4**	**527.0**	**589.0**	**589.6**	**656.3**
Which element is it from?										

Element	**Its lines (nm)**
Hydrogen	656.3 · 486.1 · 434.0 · 410.2
Helium	587.6
Lithium	670.8
Sodium	589.0 · 589.6
Magnesium	516.7 · 517.3 · 518.4
Potassium	766.5 · 769.9 · 404.4 · 404.7
Calcium	422.7
Iron	438.4 · 527.0
Strontium	460.7

1. Which elements are certainly in the Sun's atmosphere? What do you base it on?

2. Which ones do not appear in the list? Does that prove they are not in the Sun? Why?

3. Helium is named after the Sun. Why is it not in the list? Hint: how was it discovered?

Practice

1. Sodium's yellow light has a wavelength of 589 nm. Calculate the energy of each photon ($h = 6.63 \times 10^{-34}$ J·s; $c = 3.00 \times 10^{8}$ m/s).

2. Without calculating, explain which photon carries more energy, lithium's red (670.8 nm) or sodium's yellow (589 nm), and why.

3. Write 434 nm in metres, in scientific notation, and explain which rung hydrogen's electron drops from to give that line.

For those who want more: the fifth line

With hydrogen's ladder, calculate the wavelength of the photon from the jump from 7 to 2. Would you see it? And the one from the jump from 8 to 2? If the jump starts higher and higher each time, what value do the lines get closer to? Why?

Keep thinking

Comte said "never", and he was wrong. Are there questions science will never be able to answer? And how would we know that one is?

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